

Lecture Material #10

Information Security

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Randomness Testing & Analysis





Randomness Testing & Analysis

Randomness testing in cryptography evaluates whether sequences generated by

- random number generators (RNGs)
- or cryptographic algorithms (ciphers, hash functions) are indistinguishable from true random noise.

It is critical for ensuring that secrets like cryptographic keys, nonces, and initialization vectors are unpredictable.





Randomness Testing & Analysis

Null Hypothesis (H_0):

- ❖ 1st step in testing any hypothesis
- ❖ Set up such that it conveys a meaning that there exists no difference between the different samples
- ❖ E.g.: **Null Hypothesis** – there is **no significant difference**
- ❖ By using various tests of significance we either:
 - Reject the Null Hypothesis
 - (or)
 - Accept the Null Hypothesis
- ❖ **Rejecting** null hypothesis → difference is **significant**
- ❖ **Accepting** null hypothesis → difference is **not significant**





Randomness Testing & Analysis

Level of Significance, p-value:

- ❖ p-value is a **function of the observed sample results** (a statistic) that is used for testing a statistical hypothesis
- ❖ It is the **probability of null hypothesis being true**. It can accept or reject the null hypothesis based on p-value.
- ❖ Practically, $p < 0.01$ (1%) is considered highly significant
 $p < 0.05$ (5%) is considered significant.
- ❖ $P = 0.05 \rightarrow$ We may go wrong 5 out of 100 times by rejecting null hypothesis or, we can attribute significance with 95% confidence





Randomness Testing & Analysis

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38





Randomness Testing & Analysis

Frequency (Monobit) Test

Check whether the number of 0s and 1s in a bitstring are approximately equal.

$$\varepsilon = 1011010101 \quad n=10$$

$$S_n = 1 + (-1) + 1 + 1 + (-1) + 1 + (-1) + 1 + (-1) + 1 = 2.$$

Compute the test statistic $s_{obs} = \frac{|S_n|}{\sqrt{n}} = \frac{|2|}{\sqrt{10}} = .632455532.$





Randomness Testing & Analysis

Frequency (Monobit) Test

$$\text{erfc}(x) \approx 1 - \sqrt{1 - e^{-\frac{4}{\pi}x^2}}$$

$$\text{Compute } P\text{-value} = \text{erfc}\left(\frac{s_{obs}}{\sqrt{2}}\right) = \text{erfc}\left(\frac{.632455532}{\sqrt{2}}\right) = 0.527089.$$

Since $P\text{-value} \geq 0.01$, accept the sequence as random.

If the computed P-value is < 0.01 , then conclude that the sequence is non-random. Otherwise, conclude that the sequence is random.





Randomness Testing & Analysis

Frequency (Monobit) Test

You are given the following binary sequence of length $n = 20$:

$X = 1\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 1\ 1\ 0\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 1$

Use the Frequency (Monobit) Test to determine whether the sequence is random.





Randomness Testing & Analysis

Runs Test:

Checks whether transitions $0 \rightarrow 1$ or $1 \rightarrow 0$ are consistent with randomness.

Let

- n = length
- n_1 = # of ones
- n_0 = # of zeros

Expected number of runs: $E(R) = \frac{2n_0n_1}{n} + 1$





Randomness Testing & Analysis

Runs Test:

Bitstring: 11001000

- $n_1 = 3$
- $n_0 = 5$

$$E(R) = \frac{2(3)(5)}{8} + 1 = 4.75$$

Actual runs:

11 | 00 | 1 | 000 → 4 runs

Actual = 4, expected = 4.75 → Close enough → Random-like.





Randomness Testing & Analysis



Runs Test:

You are given the following binary sequence of length $n = 20$:

$X = 1\ 1\ 0\ 1\ 0\ 0\ 1\ 1\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 1\ 0$

Use the Runs Test to determine whether the sequence is random.





Randomness Testing & Analysis

Chi-Square Test for Uniform Distribution

Given frequencies of 0s and 1s:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Bitstring of length 100:

- Observed 1s = 58
- Observed 0s = 42
- Expected = 50 each

$$\chi^2 = \frac{(58 - 50)^2}{50} + \frac{(42 - 50)^2}{50}$$

$$\chi^2 = \frac{64}{50} + \frac{64}{50} = 2.56$$

Chi-square threshold (df=1, 0.05 level) = 3.84

2.56 < 3.84 → **Accept as random.**





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